INCH-POUND
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17 April 2024
SUPERSEDING
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17 August 1994

DETAIL SPECIFICATION

HOISTS, CHAIN AND WIRE ROPE, PNEUMATIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers pneumatic powered chain and wire rope hoists.

1.2 Classification. Hoists are of the following classes and types as specified (see 6.2).

1.2.1 Classes.

a. Class 1 – Steel, conventional weight

b. Class 2 – Steel or aluminum alloy, light weight

1.2.2 Types.

a. Type A – Hook suspension

b. Type B – Plain trolley suspension

c. Type C – Geared trolley suspension, hand operated

d. Type D – Geared trolley suspension, pneumatic-motor operated

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

RR-W-410	-	Wire Rope and Strand
TT-C-490	-	Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-15024	-	Plates and Tags for Identification of Equipment, General Specification for
MIL-P-15024/5	-	Plates, Identification
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-S-24093	-	Steel Forgings, Carbon and Alloy, Heat Treated
MIL-DTL-24441	-	Paint, Epoxy-Polyamide, General Specification for
MIL-PRF-24508	-	Grease, High Performance, Multipurpose (Metric)
MIL-PRF-24635	-	Coating System, Weather-Resistant, Exterior Use
MIL-C-24707	-	Castings, Ferrous, General Specification for
MIL-DTL-24707/1	-	Castings, Ferrous, for Machinery and Structural Applications
MIL-C-24707/5	-	Castings, Ductile Iron and Austenitic Ductile Iron

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-889	-	Galvanic Compatibility of Electrically Conductive Materials
MIL-STD-1472	-	Human Engineering
MIL-STD-1474	-	Noise Limits
MIL-STD-2035	-	Nondestructive Testing Acceptance Criteria

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470	-	Designing and Developing Maintainable Products and Systems, Volume I
MIL-HDBK-781	-	Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels
T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 6013 - Standard for Industrial Enclosed Gear Drives

AGMA 6034 - Practice for Enclosed Cylindrical Wormgear Speed Reducers and Gearmotors

(Copies of these documents are available online at www.agma.org.)

ASSOCIATION FOR MATERIALS PROTECTION AND PERFORMANCE (AMPP)

SSPC-SP 10 - Near White Blast Cleaning

(Copies of this document are available online at www.ampp.org.)

ASTM INTERNATIONAL

ASTM A48/A48M - Standard Specification for Gray Iron Castings

ASTM A143/A143M - Standard Practice for Safeguarding Against Embrittlement of Hot-Dip Galvanized Structural Steel Products and Procedure for Detecting Embrittlement

ASTM A304 - Standard Specification for Carbon and Alloy Steel Bars Subject to End-Quench Hardenability Requirements

ASTM A322 - Standard Specification for Steel Bars, Alloy, Standard Grades

ASTM A576 - Standard Specification for Steel Bars, Carbon, Hot-Wrought, Special Quality

ASTM B26/B26M - Standard Specification for Aluminum-Alloy Sand Castings

ASTM B633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60529 - Degrees of Protection Provided by Enclosures (IP Code)

(Copies of this document are available online at www.iec.ch.)

SAE INTERNATIONAL

SAE AMS-A-21180 - Aluminum-Alloy Castings, High Strength

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.2.

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3.2 Materials. Materials shall be as specified herein. Components that are in or supported by the Federal supply system shall be used to the fullest extent possible. Materials shall possess the properties necessary to withstand vibration, shock, and other conditions specified herein. Materials for use in exposed locations subject to salt water spray shall be free from all defects and imperfections that may affect serviceability or appearance of the hoist.

3.2.1 Metals. Metals used in the construction of the hoists for exposed locations shall be, to the greatest extent possible, selected from the following:

a. Steel alloys up to yield strength levels of 130,000 pounds per square inch (psi).

b. Aluminum alloys 5456 and 5086, strain hardened to H-116.

c. Aluminum alloy castings in accordance with SAE AMS-A-21180 or ASTM B26/B26M.

d. Corrosion-resisting steel alloys 316 and 316L.

e. Stainless steel alloy 17-4PH, heat treated to a maximum yield strength of 130,000 psi.

f. Nickel-copper alloys.

g. Nickel-iron alloys.

h. Steel castings in accordance with MIL-C-24707 and MIL-DTL-24707/1.

i. Steel forgings in accordance with MIL-S-24093.

When specified (see 6.2), permission for the use of metals or structural materials other than those listed may be granted.

3.2.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.3 Prohibited materials. The hoist shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs.

a. Asbestos. Asbestos shall not be used.

b. Cadmium. Cadmium shall not be used.

c. Beryllium. Beryllium shall not be used.

d. Pressed steel. Pressed steel shall not be used except as specified for a particular application (see 6.2).

e. Magnesium. Magnesium or magnesium-based alloys shall not be used. This does not include steel or aluminum alloys that contain less than 0.5 percent magnesium.

f. Brittle material. Brittle material (see 6.7.1) shall not be used except as specified for a particular application (see 6.2).

g. Cast iron. Cast iron in any form shall not be used except where permitted by referenced specifications. The use of cast iron is limited to those alloys conforming to ASTM A48/A48M, class 35 or better. Cast nodular graphitic iron or malleable iron conforming to MIL-C-24707/5 is acceptable for machinery equipment components.

3.2.4 Dissimilar metals. Coupled dissimilar metals shall be protected against corrosion in a marine environment in accordance with MIL-STD-889.

3.3 Safety. Hoists shall meet the human safety interface requirements of MIL-DTL-1472. Equipment design shall consider the following safety criteria at a minimum:

a. Avoiding or eliminating identified hazards by design selection or material selection.

b. Controlling and minimizing hazards to personnel, equipment, and materials that cannot be avoided or eliminated.

c. Isolating hazardous substances, parts, and operations from other activities, areas, personnel, and incompatible materials.

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d. Incorporating fail-safe design features so that failures will not cause injury to personnel or damage to equipment.

e. Locating equipment parts so that access to them by personnel during operation, maintenance, repair, or adjustment shall not require exposure to hazards such as chemical burns, electrical shock, sharp edges or points, or toxic atmospheres.

f. Providing warning and caution notes in operations, assembly, maintenance, and repair instructions, and distinctive warnings of hazards on the equipment.

3.4 Design. Hoists shall be designed in accordance with the requirements specified herein.

3.4.1 Design philosophy. The design philosophy for the hoists is to achieve the performance, safety, maintainability, and reliability specified herein. The weight of the system shall be kept to a minimum (see 3.4.12, 3.4.12.1, 3.4.12.2, and 3.4.12.3) while meeting the requirements in this section. The system shall be designed to reduce maintenance requirements (see 3.7.3).

3.4.2 Strength. Components shall have the strength to lift the required load (see 3.4.12) and operate satisfactorily in accordance with the requirements specified herein. Parts shall be as light and compact as practicable and consistent with the required strength (see tables I, II, and III).

3.4.3 Moderate sea conditions. The hoist shall operate with the rated load and at rated speed under moderate sea conditions and loading factors or motion conditions as specified (see 6.2).

3.4.4 Temperature. Hoists shall operate through a temperature range of -40 through +140 °F for not less than 3,000 cycles (see 6.7.2) without a failure.

3.4.5 Service life. Moving parts, except bearings, shall have a life in rated service of not less than 10,000 hours and shall operate for not less than 1,200 hours without adjustment or replacement of parts. Periodic adjustment of brakes due to lining or friction disc wear is expected.

3.4.6 Fail-safe operation. Hoists shall be constructed for fail-safe operation. Failure of the air source or manual drive mechanism shall not result in damage to the load, jeopardize the safety of personnel, or result in uncontrolled movement of the equipment and load. In the event of a loss of air, the hoist shall come to a controlled stop without damage. Restoration of air shall not result in automatic resumption of any equipment operation that may jeopardize the safety of the load, equipment, or personnel. Fail-safe requirements shall not be applied to static components, structural members, or other static parts of mechanisms.

3.4.7 Stress. Hoist mechanical components shall use the loading factors specified in 3.4.7.1. The maximum combined stresses in component parts shall not exceed 35 percent of the tensile yield strength of the material used for hoist operation at rated load under the 15-degree incline and tilt condition. The combined stresses in component parts shall be not greater than 70 percent of the tensile yield strength of materials when the hoist is subjected to static or dynamic load tests (see 4.5.9).

3.4.7.1 Loading factor. Hoist components shall incorporate the following loading factors:

a. Dynamic loading based on inertial forces – 1.5 times rated load.

b. Static loading based on inertial forces – 2.0 times rated load.

3.4.8 Shock. When specified (see 6.2), the hoist, in the unloaded and stowed (not operating) condition, shall withstand the shock test for grade A or grade B equipment in accordance with MIL-DTL-901 (see 4.5.10).

3.4.9 Mechanical vibration. Unloaded hoists shall conform to the vibration requirements of MIL-STD-167-1, type I (see 4.5.11).

3.4.10 Noise. Each hoist shall be properly adjusted, balanced, and free from excessive noise during operation. Airborne noise levels shall be kept to a minimum and shall be not greater than the maximum level of grade D in accordance with MIL-STD-1474.

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3.4.11 General construction. Hoists shall be composed primarily of a frame or housing, a pneumatic motor (if not manually operated), a gear train to the wire rope drum or chain sprocket, a brake, an air valve control, a wire rope drum or chain sprocket, a load block, a wire rope or chain, a bottom hook, and a top hook or trolley. Nuts, bolts, collars, and lock rings shall be provided to prevent any component working loose. Hoists shall be complete in all respects including operating controls and other equipment necessary for operation. Hoist parts shall be readily accessible for servicing and replacement as required. The hoist shall not be provided with an overload limiting device or load limiter.

3.4.12 Hoist characteristics. For a specified rated load (see 6.2), lift, weight, headroom (see 6.7.4), and lifting speed shall be in accordance with tables I, II, and III unless otherwise specified (see 6.2). Envelope dimensions shall be as specified (see 6.2).

3.4.12.1 Type A hoist, hook suspension. Unless otherwise specified (see 6.2), type A wire rope and chain hoists shall be in accordance with table I and the requirements specified herein.

TABLE I. Hoist type A, hook suspension.

Rated load (tons) [1/]	Standard lift [2/] (feet) (minimum)	Headroom [3/] (inches) (maximum)	Weight of hoist [2/] (pounds) (maximum)			Lifting speed [2/] (feet per minute) (minimum)
			Chain hoists		Wire rope hoists	
			Class 1	Class 2	Class 1	
¼	8	14.5	68	48	275	40
½	8	15	68	48	285	30
1	8	18	100	61	375	19
1½	8	23.5	155	107	435	15
2	8	23.5	233	107	510	10
3	8	32	270	130	550	10
4	8	37	320	138	720	8
5	8	45	413	172	890	8
6	8	45	420	195	1,000	8
8	8	49	500	305	1,230	6
10	8	54	620	322	1,400	4
12	8	54	875	350	--	4
16	8	60	1,120	600	--	4
20	8	71	1,300	1,100	--	4

NOTES:

1/ 2,000 pounds per ton.

2/ See 6.2.

3/ Maximum headroom (distance between hooks) measured from inside the hooks when the load chain is retracted to its shortest length.

3.4.12.2 Type B and type C hoists, plain and geared trolley suspension. Unless otherwise specified (see 6.2), type B and type C chain hoists shall be in accordance with the requirements of table II and the requirements specified herein. Type B and type C hoists shall meet these requirements when tested as specified in 4.5.14.2 (see 4.5.14.2.1 for plain geared trolleys and 4.5.14.2.2 for type C geared trolleys).

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TABLE II. Type B and C chain hoists, plain and geared trolley suspension.

Rated load (tons) [1/]	Standard lift [2/] (feet) (minimum)	Lifting speed [2/] (feet per minute) (minimum)	Standard size of I-beam (inches)	Headroom [3/] (inches) (maximum)	Pull to traverse hoist (pounds) (maximum)		Radius of track curve (inches) (minimum)	Chain hoist weight less track clamp (pounds) [2/] (maximum)			
					Plain trolley [4/] type B	Geared trolley [5/] type C		Type B		Type C	
								Class 1	Class 2	Class 1	Class 2
¼	8	40	5	14.5	15	--	21	137	118	275	--
½	8	30	5	15	20	5	21	137	118	275	--
1	8	10	6	18	40	10	21	240	170	284	205
1½	8	15	7	19.5	45	13	36	322	270	335	270
2	8	10	8	19.5	60	15	36	456	328	500	360
3	8	10	10	26.5	65	21	48	560	430	525	480
4	8	8	10	26.5	70	23	66	765	534	900	630
5	8	8	12	32	75	28	66	1,080	730	1,100	800
6	8	8	12	32	100	35	66	1,090	750	1,120	800

NOTES:

[1/] 2,000 pounds per ton.

[2/] See 6.2.

[3/] Maximum headroom (distance between load hook and underside of I-beam) measured from inside of load hook when load chain is retracted to its shortest length.

[4/] Direct pull on trolley (along direction of track when moving on straight, level track).

[5/] Pull on gear trolley hand chain (when moving on straight, level track).

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3.4.12.3 Type D hoists, geared trolley suspension, pneumatic power operated. Unless otherwise specified (see 6.2), type D hoists shall be in accordance with table III and the requirements specified herein. Type D hoists shall be equipped with a traversing pneumatic motor and controls for traversing the trolley. The pneumatic motor shall be capable of traversing the loaded hoist along a trolley track configured as specified in 4.5.12. Testing requirements are as specified in 4.5.14.3.

TABLE III. Type D chain hoist, geared trolley suspension, pneumatic powered.

Rated load (tons) 1/	Weight (pounds) (maximum)		Standard lift 2/ (feet)		Lifting speed 2/ (feet per min) (minimum) 3/		Headroom (maximum) 4/	
	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2
1	280	225	12	8	40	30	35	20
2	475	225	12	8	20	15	40	20
3	660	300	12	8	14	10	43	27.5
4	860	365	12	8	12	8	46	30
5	1,050	550	12	8	10	8	49	32
10	2,000	750	12	8	4	4	66	48

NOTES:

1/ 2,000 pounds per ton.

2/ See 6.2.

3/ Minimum lift speed for fully opened control valve.

4/ Maximum headroom (distance between load hook and underside of I-beam) measured from inside of load hook when load chain is retracted to its shortest length.

3.4.13 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.4.14 Housing. The housing shall be constructed of steel or aluminum alloy for maximum strength and minimum deflection. The housing shall contain the hoist mechanism, including gears, pneumatic motor, wire rope drum or chain sprocket, brake, and pneumatic controls (except when pendant control is specified [see 3.4.20.1]). These components shall be compactly and securely mounted.

3.4.15 Hoist drives. Hoist lift and type D (see 3.4.12.3) powered trolley drives shall be powered by a reversible pneumatic motor of enclosed construction that shall operate with air gauge pressures between 80 and 100 lb/in^2. The pneumatic motor shall be provided with the power and starting torque levels to operate the hoist drive for the required load (see 3.4.12) and shall operate without perceptible vibration at any of the hoist loads or speeds within the rated load and speed capacity.

3.4.15.1 Hoist lift. The hoist lift pneumatic drive motor shall be coupled through a gear reducer or drive gear to the load chain sprocket or wire rope drum.

3.4.15.2 Hoist trolley. The hoist trolley pneumatic drive motor shall be coupled through drive gears to the trolley drive wheels, friction wheel, or positive drive sprocket wheel. For positive drive units, the motor shall be coupled through a speed reducer to a chain sprocket used with a 1-inch pitch, single strand, No. 80-1 RC-A standard roller chain welded to the bottom of trolley track.

3.4.16 Hoist load lifting medium. As specified (see 6.2), link chain, roller chain, or wire rope shall be provided as the hoist load lifting medium. The chain or wire rope shall be of sufficient strength to withstand the tests specified (see 4.5) and shall meet the safety factor specified in 3.4.16.1, 3.4.16.2, or 3.4.16.3.

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3.4.16.1 Link chain. When specified (see 6.2), the hoist load lifting medium shall be link chain. The link chain shall be manufactured from an alloy steel in accordance with any of the American Iron and Steel Institute (AISI) grade designations of ASTM A304 or ASTM A322, electric or forge welded. Each chain link shall be of uniform size and shape to prevent twisting and shall seat properly in the hoist chain sprocket pockets. The chain shall provide a safety factor of not less than 5 for the hoist rated load based on the ultimate strength of the material. When specified (see 6.2), the chain shall be protected by a zinc coating in accordance with ASTM B633, type VI, class Fe/Zn 12. The safeguarding against and procedure for detecting embrittlement of zinc coating shall be in accordance with ASTM A143/A143M. The chain shall be securely attached to the hoist and easily removed.

3.4.16.1.1 Hoist load link chain sprocket. Hoist load link chain sprockets shall be provided with chain link pockets accurately shaped to fit the links of the chain, which shall operate freely and smoothly over the hoist load link chain sprocket. The hoist load link chain sprocket shaft shall be either integral with or rigidly connected to the hoist load link chain sprocket. Welding of the hoist load link chain sprocket to the shaft is not acceptable.

3.4.16.2 Roller chain. When specified (see 6.2), the hoist load lifting medium shall be roller chain. The roller chain shall be manufactured from an alloy steel. Each roller chain link shall be of uniform size and shape and shall seat properly in the hoist chain sprocket. The roller chain shall provide a safety factor of not less than 5 for the rated load based on the ultimate strength of the material. The chain shall be securely attached to the hoist and easily removed.

3.4.16.2.1 Hoist load roller chain sprockets. Hoist load roller chain sprocket teeth shall mate with the spacing of the roller chain rollers, which shall operate freely and smoothly over the sprocket. The hoist load roller chain sprocket shall be rigidly connected to the hoist load roller chain shaft. Welding of the hoist load roller chain sprocket to the shaft is not acceptable.

3.4.16.3 Wire rope. When specified (see 6.2), the hoist load lifting medium shall be wire rope. Wire rope shall be in accordance with RR-W-410. The wire rope shall provide a safety factor of not less than 5 for the rated load based on the minimum breaking strength of the wire rope.

3.4.16.3.1 Hoist load wire rope drums. Hoist load wire rope drums shall have machined radiused grooves. Wire rope drum grooves shall have a minimum depth equal to 40 percent of the wire rope diameter. Wire rope drums shall be fitted on each end with recessed flanges to prevent wire rope jamming. The wire rope drum diameter shall be not less than 20 times the diameter of the wire rope except that, when extra flexible wire rope is used, the diameter of the drum may be 14 times the diameter of the wire rope. There shall be not less than 1½ full turns of wire rope remaining on the drum with the hook in the lowest elevation of the rated lift. Wire rope drums shall have a rope winding guide and shall be protected on both the top and sides. The wire rope shall be securely attached to the wire rope drums.

3.4.16.4 Hoist load chain stowage. Hoist construction shall include means for stowing the full length of load chain in chain reels, bags, or baskets as specified (see 6.2) when the hook is in the up position. The chain reels shall provide for stowage of chain for 8 feet of lift. The bags or baskets shall be used for stowage of chain for lifts greater than 8 feet and shall be located to prevent interference with the low headroom features of the hoist (see 3.4.12). The chain reels shall maintain a constant tension force to prevent chain slack between the load sprocket and the chain stowage. The load sprocket and load chain stowage shall provide a constant chain feed without binding or jamming in the chain guide, stowage, or hoist frame.

3.4.16.5 Chain guides. Enclosed chain guides shall ensure the hoist load roller or link chain enters the sprocket in the proper position to prevent misalignment of the hoist load chain and sprocket. The guides, if bolted on, shall not loosen under vibration.

3.4.17 Load hooks. The hoist load and hoist suspension hooks shall be of drop forged, heat-treated steel that shall withstand the test loads specified in 4.5.9. The hoist load and hoist suspension hooks shall be as shown on figure 1, as specified (see 6.2), and shall be fitted with a swivel and a safety device, except type I, fixed eye, regular hooks. The safety device shall consist of a spring latch or swivel type closure to bridge the hook throat opening and shall be an integral part of the hoist hook. Spring latch type closures shall be hinged at the neck of the hook with the other end resting inside the tip of the hook. The swivel shall permit 360-degree rotation of the hoist load hook with full load without twisting the attached wire rope, link, or roller chain. Hoist load and suspension hook throat openings shall be in accordance with the dimensions shown in table IV. The hook throat safety device of the suspension or upper hook shall withstand a pull against the hook throat safety device as specified (see 4.5.11).

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Hooks shall be clearly marked with manufacturer identification and allowable hook load or allowable hook load designator. Type A hoist mounting hooks shall meet these requirements when tested in accordance with 4.5.19.2.

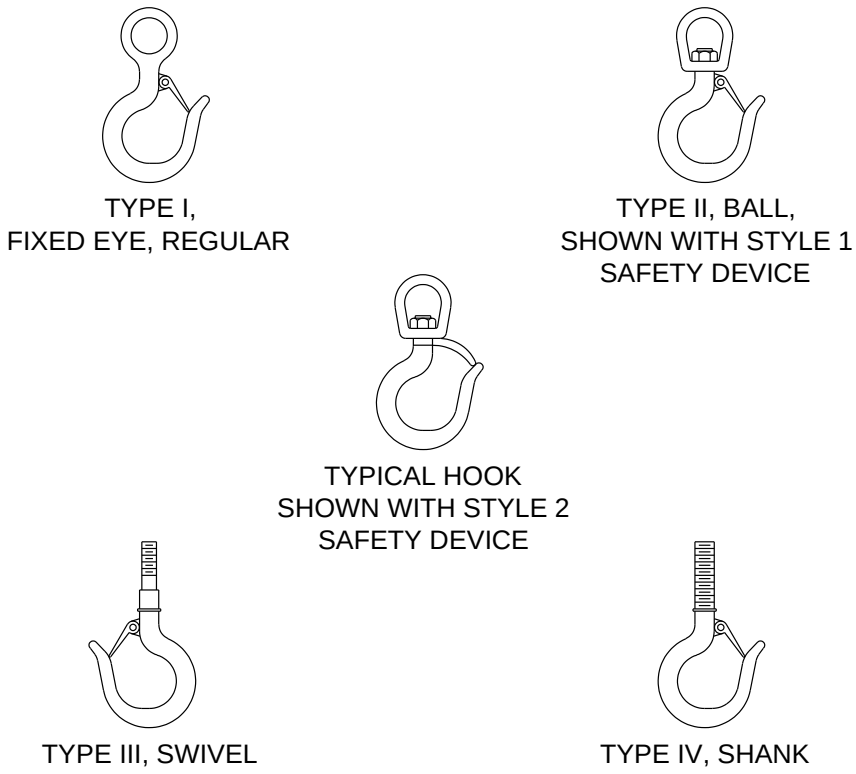


FIGURE 1. Illustrating types of hoist hooks (shown with safety devices, style 1 and style 2).

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TABLE IV. Hook throat openings.

Hoist rated load (pounds)	Hook throat opening (inches) (minimum)
1,000	0.750
2,000	0.906
3,000	1.0
4,000	1.125
5,000	1.125
6,000	1.5
7,500	1.375
10,000	1.625
11,000	2.0
13,000	2.063
15,000	2.063
17,000	2.063
20,000	2.25
25,000	2.25
30,000	2.75
40,000	3.0

3.4.17.1 Load hook finish. When specified (see 6.2), hooks shall be zinc plated. Zinc plating shall be in accordance with ASTM B633, type VI, class Fe/Zn 12. The safeguarding against and procedure for detecting embrittlement of zinc coating shall be in accordance with ASTM A143/A143M. The hook throat safety device shall be constructed of non-corrosive material or treated for corrosion resistance.

3.4.18 Range of load hook.

3.4.18.1 Chain hoists. Link and roller chain hoists shall pick up a load with the load hook anywhere within a radius of 2 feet perpendicular to the load chain sprocket centerline and 7 feet below the load sprocket without jamming or jumping the pockets or teeth of the load sprocket.

3.4.18.2 Wire rope hoists. Wire rope hoists shall pick up a load with the load hook anywhere within a radius of 2 feet perpendicular to the wire rope drum centerline at the point the wire rope reeves on the wire rope drum, and 7 feet below the load wire rope drum, without jamming or jumping the wire rope drum.

3.4.19 Load blocks.

3.4.19.1 Wire rope load block. The wire rope load block shall have a housing and a load sheave with rounded corners and a hook as specified in 3.4.17. The wire rope load block sheave shall be mounted on a steel axle and means shall be provided for ample lubrication of the wire rope load sheave and axle at all times. The wire rope load block shall prevent the wire rope from jamming during normal operations.

3.4.19.2 Chain load block. The chain load block shall be as specified in 3.4.19.1, except that it shall be constructed for use with link or roller chain instead of wire rope. The chain load block shall prevent the load chain from jamming during normal operation.

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3.4.20 Controls. The hoist pneumatic motor shall be valve-controlled either by an operating lever actuated by hand ropes or chains properly attached to the hoist or controlled by a pendant control unit. The length of the hand ropes or chains shall be enough to attach to the hoist and allow operation at all times while it is being used. The speed of the motor shall be regulated by the valve opening. The valve shall be fitted with a self-centering device that automatically restores the valve to the closed position when the hand rope or chain is released. The controls shall vertically position a load within ±0.250 inch.

3.4.20.1 Pendant control. When specified (see 6.2), a pendant control unit shall be provided in lieu of a hand rope or chain control. Thumb operated controls shall be on a hand-held control unit suspended on a pendant of specified length (see 6.2) to allow easy operation of the hoist for the required lift range. The pendant controls shall support a 200-pound load. Controls shall permit easy selection of speeds from minimum to maximum speeds when handling rated load, as well as metering of air supply to the motors for vernier positioning of the load in very limited space. "UP" and "DOWN" controls shall be labeled as such. Traversing controls shall be color coded, indicate with an arrow the direction of travel, and be affixed to the hoist so as to be visible from operating positions on either side of the hoist.

3.4.21 Air supply requirements. The air supply line shall connect to the hoist and not to the pendant control.

3.4.21.1 Air supply characteristics. The hoist shall be capable of operating with an air supply having the following characteristics:

a. A moisture content of not greater than 0.002 pound of water per pound of dry air at 60 °F and 90 lb/in^2 absolute.

b. Solid particle contamination limited to 25 micrometers.

c. Not less than one drop of atomized lubrication for every 10 cubic feet per minute (cfm) of air.

3.4.22 Hoist brake. The hoist brake shall be spring loaded and of the automatic operating type, which shall stop hoist motion when the pneumatic pressure is reduced below the safe motor operating pressure. The hoist brake shall be self-adjusting or readily accessible for easy adjustment to compensate for wear of the brake lining. The hoist brake shall hold the test loads required from a stopped position and shall stop and hold rated loads without slipping (see 4.5.9). The brake shall be equipped with a manual release for use in the event of a loss of air pressure. Manual release mechanisms shall be arranged so that they can be operated without endangering the operator.

3.4.23 Gears. Gears shall be spur, helical, or worm and wheel type manufactured in accordance with AGMA 6013 and AGMA 6034. Gears shall be totally enclosed in a readily accessible casing that will permit examination, servicing, and cleaning. Gears shall have an ingress protection (IP) rating of IP64 in accordance with IEC 60529.

3.4.24 Bearings. Bearings shall be sealed or shielded anti-friction bearings that have a service life of not less than 10,000 hours. Bearings shall have load ratings greater than the maximum bearing loads required for the hoists under the operating conditions specified herein. If bearings are not permanently lubricated as specified, they shall be provided with fittings to permit lubrication. Where practicable, rotating shafts shall be supported in anti-friction bearings or bushings, or both, and shall be enclosed against entry of foreign matter. When anti-friction bearings cannot be used to support rotating shafts, bearings or other means shall meet the design requirements of anti-friction bearings.

3.4.25 Lubrication. Means shall be provided for the lubrication of all moving parts of the hoist and trolley. Where practicable, points requiring lubrication shall be permanently lubricated. If permanent lubrication cannot be achieved, fittings shall be installed that shall be readily accessible. Lubrication shall be in accordance with MIL-PRF-24508 and MIL-PRF-17331 or manufacturer's specifications provided that lubricant is readily available and free of ozone depleting substances. Provisions shall be made for lubricating all rotating and sliding surfaces or they shall be provided with permanent lubrication.

3.4.26 Limit stops. Upper and lower limit stops shall be installed to automatically stop the hoist in the maximum lifting or lowering positions to prevent the load hook from overtravel.

3.4.27 Manual operation. When specified (see 6.2), means such as a hand chain wheel or hand-crank shall be installed on the hoist for raising, lowering, traversing, and track clamp operation. To provide hand clearance for operator safety, the length and location of the hand crank shall provide for a minimum of 1 inch of operational hand clearance measured vertically between the hand crank and the top of the smallest specified I-beam trolley track,

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track foundation, or hull structure. The force required on a crank to lower the rated load shall not exceed 40 pounds-force (lbf). The load shall not lower unless the brakes are intentionally and manually released, or the hand crank is manually cranked.

3.4.27.1 Manual operation hand chain wheels. When manual operation hand chain wheels are used, they shall be made of material compatible with the hand chain used and shall have pockets shaped to fit the links of the hand chain as specified (see 6.2). The hand chain wheels shall be equipped with a chain guide that shall permit operation of the hand chain from an angle of 10 degrees out from either side of the hand chain wheel without slipping or jumping off the hand chain wheel.

3.4.27.2 Manual operation hand chain. Chain for manual operation chain wheels shall be link chain manufactured from electric or forge welded carbon steel in accordance with ASTM A576 or alloy steel in accordance with any of the AISI grade designations of ASTM A304 or ASTM A322. Unless otherwise specified (see 6.2), hand chains shall be endless link chain and shall have a deck clearance of not less than 2 feet. Hand chains shall have a breaking strength of not less than three times the maximum pull required to lift the maximum rated load. Each chain link shall be of uniform size and shape to prevent twisting and shall seat properly in the hoist chain sprocket. Enclosed chain guides shall be provided and shall prevent the chain from jumping the sprocket. When specified (see 6.2), the chain shall be protected by a zinc coating in accordance with ASTM B633, type VI, class Fe/Zn 12. The safeguarding against and procedure for detecting embrittlement of zinc coating shall be in accordance with ASTM A143/A143M.

3.4.27.3 Release mechanism. A release mechanism shall be installed to allow manual lowering in the event of a loss of pneumatic power. An interlock shall be installed to prevent pneumatic operation of the hoist during manual operation.

3.4.28 Trolley suspension hoists, types B, C, and D.

3.4.28.1 Trolley wheels. Trolleys shall be constructed with not less than four wheels. Trolley wheels shall be of ferrous material. Trolley wheels shall have treads hardened to not less than 285 Brinell hardness number (Bhn). Trolleys up to and including 3-ton capacity shall have wheels of solid or pressed steel with treads hardened to a depth of not less than 0.020 inch. For trolleys over 3-ton capacity, wheels shall be forged or solid with treads hardened to a depth of not less than 0.063 inch. Trolley wheels shall be concentric within 0.030 inch and cast wheels shall have machine treads. Where the trolley wheel mounting arrangement is such that the wheels are not canted to the flange of the I-beam type trolley track (tapered treads), wheel treads shall be convex in shape to minimize wear. The operating device for geared trolleys shall be either a manual chain wheel equipped with an endless chain of material and length (see 6.2) as specified in 3.4.27, or a pneumatic motor-powered drive system as specified in 3.4.15. Unless otherwise specified (see 6.2), trolley wheel spacing shall be designed for use on applicable standard I-beam flange sizes.

3.4.28.2 Trolley wheel bearings and axles. Trolley wheels shall be equipped with bearings as specified in 3.4.24. The wheel bearings shall be protected from foreign materials. Trolley wheel axles shall be made from alloy steel.

3.4.28.3 Trolley wheel gear and drive pinion. The trolley wheel gear and drive pinion shall be made of steel or malleable iron.

3.4.28.4 Trolley brake. The trolley power unit shall be equipped with a brake coupled to the drive transmission. The brake shall be of the spring loaded automatic operating type that shall actuate upon reduction of pneumatic pressure below the safe motor operating pressure. The brake shall be equipped with a manual release for emergency operation. The manual release mechanism shall be arranged so that it can be operated without endangering the operator. The brake shall hold the test loads without slipping from a stopped position with rated load on the hoist as specified (see 4.5.12). The brake shall be adjustable.

3.4.28.5 Trolley equalizers. Means shall be provided for distributing the hoist load equally into the trolley side frame (sideplates).

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3.4.28.6 Trolley hoist track clamps. When specified (see 6.2), the trolley containing the suspension hoists, which shall be fully loaded, shall be locked to the track by quick-acting track clamps. The clamps shall be adjustable for wear and function equally well on curved or straight track. The clamps shall function without increasing the trolley wheel shaft or wheel bearing load and in such a manner that the stresses resulting from locking will be taken up in the trolley frame. The operating device for track clamps shall be either a chain wheel equipped with an endless chain of material and length as specified in 3.4.27, a lever, or a drum and lanyard. The hand pull required to set or release the trolley track clamp shall be not greater than 80 pounds. The chain or lanyard drop from the trolley support beam shall be approximately 2 feet less than the specified lift of the hoist (see tables II and III). Track clamps shall show no sign of slipping or permanent deformation when tested as specified (see 4.5.6).

3.4.28.7 Trolley for curved track. Unless otherwise specified (see 6.2), trolley wheels shall be spaced to negotiate curved track at the radii specified (see table II).

3.4.28.8 Trolley wheel preventer bars. Trolley wheel flanges shall be prevented from riding up onto the supporting beam by means of preventer bars.

3.4.29 Painting. Exposed surfaces, except hooks, chains, and other wear surfaces, shall be cleaned, pretreated, and finish coated in accordance with the requirements herein. Paints and coatings shall be lead-free and chromate-free. Exposed surfaces, except hooks, chains, and other wear surfaces, shall be painted as specified herein. Prepare surfaces to be painted by abrasive blasting to SSPC-SP 10 near-white metal condition and establish a surface profile of 2 to 4 mils; if abrasive blasting to SSPC-SP 10 is not possible, then new bare steel housing shall be chemically cleaned and pre-treated with a zinc or iron phosphate treatment per TT-C-490. The first coat of paint applied shall be MIL-DTL-24441, type III, formula 159 zinc-rich epoxy at a thickness of 3 to 4 mils dry film thickness (DFT). The second coat of paint applied shall be MIL-DTL-24441, type IV, formula 150 green epoxy at a thickness of 4 to 6 mils DFT. The top coat of paint, with the color as specified (see 6.2), applied shall be a MIL-PRF-24635, type V or VI polysiloxane coating at a thickness of 5 to 8 mils DFT. All manufacturer's instructions shall be followed regarding application conditions and procedures (e.g., temperatures, humidity, pot life, cure times, etc.).

3.5 Identification plate. An identification plate shall be installed in a prominent location on each hoist in accordance with MIL-DTL-15024 and MIL-P-15024/5. The plate shall identify the following:

a. Weight and shock (grade), as applicable
b. Rated load and hoisting speed
c. Manufacturer's name and part, serial, or model number
d. Military specification (MIL-DTL-2813)
e. Class and type
f. Contract or order number
g. Date of manufacture
h. National Stock Number (NSN) (if established)

3.6 Operating instructions. The operating instructions manual shall be placed in a waterproof envelope marked "Operating Instructions". The envelope shall be attached securely to the hoist.

3.7 Availability, reliability, and maintainability.

3.7.1 Availability. The minimum acceptable inherent availability (Ai) of the hoist shall be 0.90. This requirement establishes threshold values for reliability, maintainability, and supportability of the hoist. Availability is defined in MIL-HDBK-470.

3.7.2 Reliability. The hoist shall operate for an average period of 3,000 continuous cycles without failure. This value of 3,000 mean cycles between failure (MCBF) is equivalent to 90 days of normal ship operation without hoist failure. Reliability is defined in MIL-HDBK-781.

3.7.3 Maintainability. Routine corrective maintenance at the organizational level shall be accomplished by replacing complete assemblies and subassemblies. The mean time to repair (MTTR) for the hoist shall be 4 hours. Not less than 95 percent of all corrective maintenance actions shall require not greater than 10 hours to complete. Maintainability is defined in MIL-HDBK-781.

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3.8 Workmanship. The hoist shall withstand any operational requirements specified herein without permanent deformation, malfunction, or component failure or interference caused by improper workmanship. Parts of the hoist, before and after painting, shall be clean and free of sand, rust, dirt, fins, pits, sprues, scales, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded so that a hazardous surface does not exist. Bolted connections shall use standard bolts and self-locking nuts. Threads shall have full thread engagement.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on pneumatic powered chain and wire rope hoists in accordance with table V when a first article sample is required (see 3.1). One first article sample successfully examined and tested for acceptability of operating characteristics as specified in 4.4 and 4.5 shall serve as a manufacturing standard and be used as a guide in the inspection of production hoists produced under the contract. If required (see 6.2), one other first article model shall be tested in accordance with MIL-DTL-901 (see 3.4.8 and 4.5.10).

TABLE V. First article and conformance inspections.

Characteristic	First article	Conformance	Requirement	Verification
Materials	X	X	3.2	4.5.1
Safety criteria	X	--	3.3	4.5.2
Weight of system	X	--	3.4.1	4.5.3
Strength	X	--	3.4.2	4.5.4
Sea state conditions	X	--	3.4.3	4.5.5
Temperature	X	--	3.4.4	4.5.6
Service life	X	--	3.4.5	4.5.7
Fail-safe operation	X	X	3.4.6	4.5.8
Stress	X	--	3.4.7	4.5.9
Loading factor	X	X	3.4.7.1	4.5.9.1, 4.5.9.2
Shock	X	--	3.4.8	4.5.10
Mechanical vibration	X	--	3.4.9	4.5.11
Noise	X	X	3.4.10	4.5.12
Construction	X	X	3.4.11	4.5.13
Hoist characteristics	X	X	3.4.12	4.5.14
Type A hoist	X	X	3.4.12.1	4.5.14.1
Type B and C hoists	X	X	3.4.12.2	4.5.14.2
Plain trolley test	X	X	3.4.12.2	4.5.14.2.1
Geared trolley test	X	X	3.4.12.2	4.5.14.2.2
Type D hoist	X	X	3.4.12.3	4.5.14.3
Operating tests	X	X	3.4.13	4.5.14.4
Interchangeability	X	--	3.4.13	4.5.15
Housing	X	X	3.4.14	4.5.16

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TABLE V. First article and conformance inspections - Continued.

Characteristic	First article	Conformance	Requirement	Verification
Hoist drives	X	--	3.4.15	4.5.17
Hoist lift	X	--	3.4.15.1	4.5.17.1
Hoist trolley	X	X	3.4.15.2	4.5.17.2
Hoist load lifting medium	X	--	3.4.16	4.5.18
Link chain	X	--	3.4.16.1	4.5.18.1
Hoist load link chain sprocket	X	X	3.4.16.1.1	4.5.18.1.1
Roller chain	X	--	3.4.16.2	4.5.18.2
Hoist load roller chain sprocket	X	X	3.4.16.2.1	4.5.18.2.1
Wire rope	X	--	3.4.16.3	4.5.18.3
Hoist load wire rope drums	X	--	3.4.16.3.1	4.5.18.3.1
Hoist load chain stowage	X	X	3.4.16.4	4.5.18.4
Chain guides	X	X	3.4.16.5	4.5.18.5
Load hooks	X	X	3.4.17	4.5.19
Load hook finish	X	X	3.4.17.1	4.5.19.1
Mounting hook type (type A hoist)	X	X	3.4.17	4.5.19.2
Range of load hook	X	X	3.4.18	4.5.20
Load block	X	X	3.4.19.1	4.5.21
Controls	X	X	3.4.20	4.5.22
Pendant control	X	X	3.4.20.1	4.5.22.1
Air supply	X	X	3.4.21	4.5.23
Air supply characteristics	X	X	3.4.21.1	4.5.23
Hoist brake	X	X	3.4.22	4.5.24
Gears	X	--	3.4.23	4.5.25
Bearings	X	--	3.4.24	4.5.26
Lubrication	X	X	3.4.25	4.5.27
Limit stops	X	X	3.4.26	4.5.28
Manual operation	X	X	3.4.27	4.5.29
Manual operation hand chain wheels	X	X	3.4.27.1	4.5.29.1
Manual operation hand chain	X	X	3.4.27.2	4.5.29.2
Release mechanism	X	X	3.4.27.3	4.5.29.3
Geared trolley traverse test	X	X	--	4.5.29.4
Hand operation for type B and C hoists	X	X	--	4.5.29.5
Trolley wheels	X	X	3.4.28.1	4.5.30.1
Trolley wheel bearings and axles	X	X	3.4.28.2	4.5.30.2
Trolley wheel gear drive pinion	X	X	3.4.28.3	4.5.30.3
Trolley brake	X	X	3.4.28.4	4.5.30.4
Trolley equalizers	X	X	3.4.28.5	4.5.30.5
Trolley hoist track clamp	X	X	3.4.28.6	4.5.30.6

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TABLE V. First article and conformance inspections - Continued.

Characteristic	First article	Conformance	Requirement	Verification
Trolley for curved track	X	X	3.4.28.7	4.5.30.7
Trolley wheel preventer bars	X	X	3.4.28.8	4.5.30.8
Painting	X	X	3.4.29	4.5.31
Identification and instruction plates	X	X	3.5	4.5.32
Operating instructions	X	X	3.6	4.5.33
Availability	X	--	3.7.1	4.5.34.1
Reliability	X	--	3.7.2	4.5.34.2
Maintainability	X	--	3.7.3	4.5.34.3
Workmanship	X	X	3.8	4.5.35

4.3 Conformance inspection. Conformance inspection shall be performed at the frequency specified (see 6.2). One sample selected in accordance with 4.3.2 shall be subjected to conformance inspection in accordance with table V and 4.5.1 through 4.5.35, as applicable.

4.3.1 Inspection lot. For the purpose of conformance inspection and test sampling, a lot is defined as all the units of the same size, type, and class, produced in one facility, using the same production processes and materials, and being offered for delivery at one time. The sample unit shall be one complete hoist unit.

4.3.2 Sampling for visual examination and testing. As a minimum, a sample quantity of completed hoists shall be selected from each lot in accordance with tables VI and VII and inspected in accordance with 4.5. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. If a lot is rejected, 100 percent of the rejected lot may be screened for the defective characteristic(s) or a new lot may be provided, which shall be inspected in accordance with the sampling plan provided herein. Requirements for maintenance of records of inspections, tests, or any resulting rejections shall be as specified (see 6.2).

TABLE VI. Sampling for visual examination.

Lot size	Sample size	
	Major defects	Minor defects
2 to 25	5	3
26 to 50	5	5
51 to 90	7	6
91 to 150	11	7
151 to 280	13	10
281 to 500	16	11
501 to 1,200	19	15
1,201 to 3,200	23	18
3,201 to 10,000	29	32

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TABLE VII. Sampling for tests.

Lot size	Sample size
2 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1,200	19
1,201 to 3,200	23
3,201 to 10,000	29

4.3.3 Casting inspection for class 1 hoists. Steel castings for load-bearing parts of class 1 hoists shall be radiographed to ensure sound castings for the intended service and, in addition, shall be subjected to magnetic particle or liquid penetrant inspection. The acceptability of castings subject to radiographic examination shall be based upon the radiographic standards for castings in accordance with S9074-AR-GIB-010/278 for criticality level 1. The acceptance criteria for castings subject to surface inspection shall be in accordance with MIL-STD-2035.

4.4 Examination. Sample hoist units selected as specified in 4.3.2 shall be assessed to verify conformance to requirements. For sample hoist components that require visual examinations, the visual defects shall be classified as specified in table VIII.

TABLE VIII. Classification of visual defects.

Category	Defect
Major	
101	Gear casing does not protect against foreign matter or permit ready access (see 3.4.23 and 4.5.25).
102	Hoist clearance, underside of track to inside of load hook, not as specified (see 3.4.12 and 4.5.14).
103	Load chain or wire rope not securely attached to hoist as specified (see 3.4.16 and 4.5.18).
104	Chain links not uniform size (see 3.4.16.1, 3.4.16.2, 4.5.18.1, and 4.5.18.2).
105	Load chain reel or basket does not store full length of chain (see 3.4.16.4 and 4.5.18.4).
106	Chain reel does not maintain tension on chain as specified (see 3.4.16.4 and 4.5.18.4).
107	Hook throat opening not as specified (see 3.4.17 and 4.5.19).
108	Hoist hooks not equipped with safety guard, or safety guard not as specified (see 3.4.17 and 4.5.19).
109	Hoist lift range not as specified (see 3.4.18 and 4.5.20).
110	Pendant unit not marked or supported as specified (see 3.4.20.1 and 4.5.22.1).
111	Hoist components damaged or missing.
112	Lubrication fittings missing (see 3.4.25 and 4.5.27).
113	Lubrication of moving parts not provided or as specified (see 3.4.25 and 4.5.27).
114	Trolley wheel mounting arrangement not as specified (see 3.4.28.1 and 4.5.30.1).
115	Trolley wheel treads not as specified (see 3.4.28.1 and 4.5.30.1).
116	Trolley hoist not equipped with track clamps, when specified (see 3.4.28.6 and 4.5.30.6).

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TABLE VIII. Classification of visual defects - Continued.

Category	Defect
Major	
117	Trolley track clamps do not function as specified (see 3.4.28.6 and 4.5.30.6).
118	Trolley wheel preventer bars missing or do not prevent trolley wheel flanges from riding up onto the supporting track (see 3.4.28.8 and 4.5.30.8).
119	Hoist lift height not as specified (see tables I through III).
Minor	
201	Hoist weight not as specified (see 3.4.12.1, 3.4.12.2, 3.4.12.3, and 4.5.14.1, 4.5.14.2, and 4.5.14.3).
202	Hoists, hook suspension, with the hoist fully retracted, distance between hooks not as specified (see 3.4.12.1 and 4.5.14.1).
203	Chain finish not as specified (see 3.4.16.1, 3.4.27.2, 4.5.18.1, and 4.5.29.1).
204	Hoists not equipped with chain container when specified (see 3.4.16.4 and 4.5.18.4).
205	Hook finish not as specified (see 3.4.17.1 and 4.5.19.1).
206	Pneumatic supply line not connected to hoist as specified (see 3.4.16.4 and 4.5.18.4).
207	Lubrication fittings damaged (see 3.4.25 and 4.5.27).
208	Paint coating not as specified (see 3.4.29 and 4.5.31).
209	Trolley wheels, concentricity not as specified (see 3.4.28.1 and 4.5.30.1).
210	Trolley wheel bearings not protected to exclude foreign matter (see 3.4.28.2 and 4.5.30.2).
211	Trolley wheel gear and drive pinion not as specified (see 3.4.28.3 and 4.5.30.3).
212	Trolley frame (sideplates) not connected by an equalizing pin (see 3.4.28.5 and 4.5.30.5).
213	Paint not as specified; paint surface broken (see 3.4.29 and 4.5.31).
214	Hoist identification plate missing or not attached as specified (see 3.5 and 4.5.32).

4.5 Design verification. Sample units selected as specified in 4.2 and 4.3 shall be tested as specified in 4.5.1 through 4.5.35.

4.5.1 Materials. Materials shall be visually inspected and tested to verify compliance with 3.2 and its sub-sections. Conformance inspection shall be limited to checking that materials are in accordance with 3.2 through 3.2.4.

4.5.1.1 Prohibited materials. Prohibited materials shall be verified as specified (see 3.2.3 and 6.2).

4.5.2 Safety criteria. The hoist shall be visually inspected and operated to verify compliance with 3.3.

4.5.3 Weight of system. The weight of the system and analysis of the maintenance requirements shall be provided to verify compliance with 3.4.1.

4.5.4 Strength. Calculations shall be performed and measurements taken to verify compliance with 3.4.2.

4.5.5 Moderate sea conditions. A Finite Element Analysis (FEA), computer modeling and simulation, or calculations shall be performed to show that the hoist can operate with the added dynamic forces of operating in moderate sea conditions to verify compliance with 3.4.3.

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4.5.6 Temperature. Visual inspection shall be conducted to determine compliance with 3.4.4. Endurance testing shall be conducted as follows to verify compliance with 3.4.4: One sample hoist of each class and type shall be subjected to 3,000 cycles of continuous operation. In the event a failure occurs during this test, and an investigation and design study result in modification to the test unit and remaining units as specified (see 6.2), the endurance test shall be repeated. After completion of the endurance tests, the gears, chain, bearings, chain sprockets, brakes, and other wearing parts shall be examined for excessive wear (see 6.7.3). During operation of these hoists, no wear particles greater than 0.031 inch in any direction shall be generated. Life-lubricated bearings requiring lubrication, load-brake lining reduced in excess of 50 percent of useful life, and reduction of bar diameter of link chain in excess of 10 percent shall be considered excessive wear. The tests specified in 4.5.9.1 and 4.5.9.2 shall be repeated to demonstrate safe operation.

4.5.7 Service life. The hoist machinery shall be operated through 20 cycles of operation at the rated load as specified (see 6.2). The motion profile for each cycle and pauses between cycles for this test shall simulate the service life specified in 3.4.5. Allow 30 seconds between cycles to allow the air motor to cool.

4.5.8 Fail-safe operation. The fail-safe features shall be operationally tested to verify compliance with 3.4.6.

4.5.9 Stress. FEA, computer modeling and simulation, or calculations shall be conducted to verify compliance with 3.4.7 and 3.4.7.1. The hoist shall be subjected to load tests as specified in 4.5.9.1 and 4.5.9.2, during which the hoists shall lift and lower a load within their rated load capacity without jerks, jars, or vibration.

4.5.9.1 Static load test. The sample hoist shall support a static load of 2.0 times the maximum rated load for a period of 10 minutes. The load shall be suspended with the hoist load chain extended to the limit of the hoist rated lift height. This extension may be changed to not less than 1 foot provided it is demonstrated that the entire length of chain supports 200 percent of rated load. The suspended test load shall be held by the hoist brake for 10 minutes. There shall be no permanent deformation of any portion of the hoist frame or load support or evidence of chain failure.

4.5.9.2 Dynamic load test. The hoist shall be loaded to 1.5 times the rated load and operated by hoisting and lowering the test load through the full operating range for ten cycles. Trolley-type hoists shall be operated back and forth over a section of track 8 feet or more in length with 150 percent load for ten cycles. The hoist and trolley shall operate satisfactorily, and the brake shall exhibit no sign of slippage. There shall be no permanent deformation of any portion of the hoist frame or load support or evidence of chain failure.

4.5.10 Shock. When specified (see 6.2), shock tests shall be conducted in accordance with the requirements of MIL-DTL-901 for grade A or grade B equipment and the hoist shall meet the requirements of 3.4.8. Shock qualification shall be certified on the applicable hoist testing placard. Testing shall be conducted on one complete unloaded hoist of each type and class. Following successful completion of the shock test, the hoist shall be subjected to the following tests:

a. Grade A - Static and dynamic tests (see 4.5.9.1 and 4.5.9.2).

b. Grade A - Plain and geared trolley tests (see 4.5.14.2.1 and 4.5.14.2.2).

c. Grades A and B - Track clamp test (see 4.5.30.6).

4.5.11 Mechanical vibration. The hoist shall be vibration tested in accordance with MIL-STD-167, type I to verify compliance with 3.4.9.

4.5.12 Noise. The hoist shall be tested for noise during operation in accordance with MIL-STD-1474 to verify compliance with 3.4.10.

4.5.13 Construction. A visual inspection and testing shall be performed to verify that the hoist was constructed in compliance with 3.4.11.

4.5.14 Hoist characteristics. A visual inspection and testing shall be performed to verify that the hoist was constructed to meet the rated load, lift, weight, headroom, envelope dimensions, and lifting speed in compliance with 3.4.12. Speed, load positioning, and performance testing shall be in accordance with 4.5.14.4 and its sub-sections.

4.5.14.1 Type A hoists. A visual inspection and testing shall be performed to verify that the type A hoist was constructed in compliance with 3.4.12.1.

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4.5.14.2 Type B or C hoists. A visual inspection and testing shall be performed to verify that the type B or C hoist was constructed in compliance with 3.4.12.2.

4.5.14.2.1 Plain trolley test. The pull required to move the rated capacity loaded plain trolley suspension hoist along a straight portion of track shall be determined by attaching a wire rope or cord to the trolley so that the pull is exerted parallel to the track and reeved over a sheave suspended from the track not less than 5 feet from the trolley. The required pull shall be measured by means of either weights or a spring balance attached to the wire rope or cord. Failure to conform to the requirements for maximum pull to traverse the hoist (see table II) shall constitute failure of this test.

4.5.14.2.2 Geared trolley test. The pull required on the type C geared trolley suspension hoist hand chain to move a hoist supporting its rated load along a straight horizontal portion of track shall be determined by attaching weights a spring balance to the hand chain. Failure to conform to the requirements for maximum pull to traverse the hoist (see table II) shall constitute failure of this test.

4.5.14.3 Type D hoists. A visual inspection shall be performed to verify that the type D hoist was constructed in compliance with 3.4.12.3. Testing shall be performed in accordance with 4.5.14.3.1 through 4.5.14.3.3 to verify compliance:

4.5.14.3.1 Traversing rated speed test. The power-operated trolley shall traverse the hoist with rated load along a horizontal trolley track to demonstrate conformance to the traversing rated speed as specified.

4.5.14.3.2 Trolley drive test (rolling conditions). The trolley shall traverse the loaded hoist on a horizontal trolley track inclined at an angle of 10 degrees to either side to demonstrate satisfactory operation. The trolley will not be required to operate at a specified speed but shall move along steadily.

4.5.14.3.3 Trolley brake test. With the hoist fully loaded and the trolley on a trolley track inclined at an angle of 15 degrees and tilted at an angle of 15 degrees, the trolley shall maintain its position on the track when stopped by means of the braking arrangement specified in 3.4.28.4.

4.5.14.4 Operating tests. The hoist shall be tested to verify that it is satisfactory for operation with the rated load as follows:

a. Hoisting rated speed test. The hoist shall be operated for approximately 90 percent of lift height to verify compliance to the hoisting rated speed requirements.

b. Lowering rated speed test. The hoist load hook shall be lowered at maximum speed to verify compliance to the lowering rated speed requirements.

c. Hoisting travel limit test. The hoist shall be operated in the lifting and lowering directions so as to engage the limit stops to demonstrate the hoist's ability to prevent load hook over-travel.

d. Load positioning sensitivity test. The hoist capability of accurately positioning a load shall be demonstrated. The test shall be conducted by establishing a reference height (approximately 12 inches above deck) and then jogging the rated load to a position ±0.250 inch above and below the reference height. This test shall be repeated not less than six times.

e. Performance test. The hoist shall be continuously operated at maximum lifting and lowering speeds through approximately 90 percent of the design lift height for not less than 30 minutes. During this test, the hoist shall operate satisfactorily without any indication of malfunction.

4.5.15 Interchangeability. A visual inspection and analysis shall be performed to verify compliance with 3.4.13.

4.5.16 Housing. A visual inspection and analysis shall be performed to verify compliance with 3.4.14.

4.5.17 Hoist drives. Analysis, calculations, and testing shall be performed to verify compliance with 3.4.15.

4.5.17.1 Hoist lift. A visual inspection and test shall be performed to verify compliance with 3.4.15.1.

4.5.17.2 Hoist trolley. A visual inspection and test shall be performed to verify compliance with 3.4.15.2.

4.5.18 Hoist load lifting medium. Analysis and calculations shall be performed to verify that the hoist load lifting medium is in compliance with 3.4.16.

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4.5.18.1 Link chain. Analysis and calculations shall be performed to verify link chain safety factors and visual inspection shall be performed to verify that the installation of the link chain is as specified in 3.4.16.1.

4.5.18.1.1 Hoist load link chain sprockets. A visual inspection and test shall be performed to verify that the hoist load link chain sprockets were correctly designed and installed in compliance with 3.4.16.1.1.

4.5.18.2 Roller chain. Analysis and calculations shall be performed to verify roller chain safety factors and visual inspection shall be performed to verify that the installation of the roller chain is as specified in 3.4.16.2.

4.5.18.2.1 Hoist load roller chain sprockets. A visual inspection and test shall be performed to verify that the hoist load roller chain sprockets were correctly designed and installed in compliance with 3.4.16.2.1.

4.5.18.3 Wire rope. Analysis and calculations shall be performed to verify wire rope safety factors are in compliance with 3.4.16.3.

4.5.18.3.1 Hoist load wire rope drums. Analysis and calculations shall be performed to verify wire rope safety factors are in compliance with 3.4.16.3.1.

4.5.18.4 Hoist load chain stowage. A visual inspection and testing shall be performed to verify load chain stowage is in compliance with 3.4.16.4.

4.5.18.5 Chain guides. A visual inspection and vibration testing shall be performed to verify compliance with 3.4.16.5.

4.5.19 Hooks. A visual inspection, analysis, and testing shall be performed to verify hook compliance with 3.4.17.

4.5.19.1 Hook finish. A visual inspection and testing shall be performed to verify hook finish compliance with 3.4.17.1.

4.5.19.2 Mounting hook test (type A hoist). The ability of the safety device of the mounting hook (see 3.4.17) to hold a load equal to the rated load of the hoist shall be tested as follows: A test load shall be attached to the closed and latched safety device in four directions. The load shall be applied to the safety device at a point measured from the hook tip along the safety device a distance equal to one-third of the throat opening as shown in table IV. The load shall first be applied alternately to opposite sides of the safety device, along the sides of the safety device at 90 degrees to the safety device in a plane perpendicular to the hook plane. The test load shall be 75 pounds for safety hoist hooks with safe working loads between 1,200 and 4,000 pounds, inclusive; 150 pounds for safe working loads between 4,000 and 10,000 pounds, inclusive; and 200 pounds for safe working loads greater than 10,000 pounds. The safety device shall suffer no permanent deformation due to the test load applications and shall be functional upon completion of testing.

4.5.20 Range of load hook. Testing shall be performed to verify compliance with 3.4.18.1 for chain hoists and 3.4.18.2 for wire rope hoists. Both type B, C, and D chain and wire rope hoists shall be tested as specified in 4.5.20.1.

4.5.20.1 Fleet angle test (type B, C, and D hoists). The hoist shall pick up a load with the hook attached to the load 2 feet out from an imaginary perpendicular 7 feet below the hoist. This test shall be completed without the chain jamming or jumping the pockets of the load sprocket. For wire rope hoists, this test shall be completed without the rope jumping the grooves on the cable drum. This test shall be conducted four times: once forward, once aft, and once on each side of the hoist.

4.5.21 Load block. A visual inspection and testing shall be performed to verify compliance with 3.4.19.1 for wire rope load blocks and 3.4.19.2 for chain load blocks.

4.5.22 Controls. Testing of the controls shall be performed to verify compliance with 3.4.20.

4.5.22.1 Pendant control. A visual inspection and testing of the pendant control shall be performed to verify compliance with 3.4.20.1.

4.5.23 Air supply. A visual inspection and testing of the air supply line shall be performed to verify compliance with 3.4.21 and its sub-sections.

4.5.24 Hoist brake. The hoist brake shall be tested to verify compliance with 3.4.22.

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4.5.25 Gears. A visual inspection shall be performed to verify compliance with 3.4.23.

4.5.26 Bearings. A visual inspection, analysis, and testing shall be performed to verify compliance with 3.4.24.

4.5.27 Lubrication. A visual inspection shall be performed to verify the lubrication is in compliance with 3.4.25.

4.5.28 Limit stops. Testing shall be performed to verify the limit stops are in compliance with 3.4.26.

4.5.29 Manual operation. The manual operation of the hoist shall be tested to verify compliance with 3.4.27 and its sub-sections. The hoist shall traverse (where applicable) and lift and lower through the full hoisting range with rated load, or lower only when manual brake release only is specified (see 3.4.27 and 3.4.28). While the hoist is being manually operated, the interlock shall prevent pneumatic operation.

4.5.29.1 Manual operation hand chain wheels. Visual inspection and testing shall be performed to verify compliance with 3.4.27.1.

4.5.29.2 Manual operation hand chain. Visual inspection, analysis or calculations, and testing shall be performed to verify compliance with 3.4.27.2.

4.5.29.3 Release mechanism. Testing shall be performed to verify compliance with 3.4.27.3.

4.5.29.4 Geared trolley traverse test. On type C geared trolley hoists, a pull of not greater than 1 pound per 200 pounds of total hoist load exerted on the hand chain shall initiate hoist movement and a pull of not greater than 1 pound per 300 pounds of total hoist load shall sustain hoist and trolley movement.

4.5.29.5 Hand operation for type B and C hoists. The hoist and the trolley of types B and C shall be operated by hand on straight and curved horizontal tracks with rated load. A pull of not greater than 1 pound (5 pounds for type B) per 200 pounds of hoist load shall be required to initiate trolley movement and a value not greater than the pull to traverse hoist value (see table II) shall be required to sustain trolley movement.

4.5.30 Trolley suspension hoists, types B, C, and D.

4.5.30.1 Trolley wheels. A visual inspection, analysis or calculations, and testing shall be performed to verify compliance with 3.4.28.1.

4.5.30.2 Trolley wheel bearings and axles. A visual inspection shall be performed to verify compliance with 3.4.28.2.

4.5.30.3 Trolley wheel gear drive pinion. A visual inspection shall be performed to verify compliance with 3.4.28.3.

4.5.30.4 Trolley brake. A visual inspection and testing shall be performed to verify compliance with 3.4.28.4.

4.5.30.5 Trolley equalizers. A visual inspection and analysis shall be performed to verify compliance with 3.4.28.5.

4.5.30.6 Trolley hoist track clamp. Visual inspection, analysis or calculations, and testing shall be performed to verify compliance with 3.4.28.6. Track clamps on type B and C hoists shall be tested by subjecting the loaded hoist to a pull equal to one-third of the rated load of the hoist. The pull shall be exerted in both directions parallel to the trolley track. The clamps shall hold the loaded hoist from moving in either direction when the trolley track is in a horizontal position.

4.5.30.7 Trolley for curved track. Testing shall be performed to verify compliance with 3.4.28.7.

4.5.30.8 Trolley wheel preventer bars. Testing shall be performed to verify compliance with 3.4.28.8.

4.5.31 Painting. A visual inspection and measurement shall be conducted to verify compliance with 3.4.29.

4.5.32 Identification plate. Identification plates shall be visually inspected to verify compliance with 3.5.

4.5.33 Operating instructions. Operating instructions shall be visually inspected to verify compliance with 3.6.

4.5.34 Availability, reliability, and maintainability.

4.5.34.1 Availability. The availability of the system shall be tested to verify compliance with 3.7.1.

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4.5.34.2 Reliability. The reliability shall be tested to verify compliance with 3.7.2 using the guidance provided in MIL-HDBK-781. The 3,000 cycles are covered in the endurance test requirements in 4.5.6.

4.5.34.3 Maintainability. The MTTR requirement shall be demonstrated by the successful completion of a maintainability demonstration using the guidance provided in MIL-HBDK-781 to verify compliance with 3.7.3.

4.5.35 Workmanship. Construction workmanship shall be visually inspected to verify compliance with 3.8.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Hoists covered by this specification are intended for general material handling by all Military activities. These hoists have design and testing requirements to meet sea state conditions for operation and corrosion prevention. The pneumatic hoists may be used to handle ammunition and explosives, when certified, cargo, boats, stores, and other shipboard items vertically within a space or between decks. Current stock of pneumatic wire rope or chain hoists procured to previous revisions of this specification may be used until depleted; future requisitions for pneumatic wire rope or chain hoists should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and class of hoist required (see 1.2).

c. When first article inspection is required (see 3.1).

d. If the use of metals or structural materials other than those listed in 3.2 is permitted (see 3.2.1).

e. Requirement for prohibited materials verification (see 3.2.3 and 4.5.1.1).

f. Applications for which pressed steel may be used (see 3.2.3.d).

g. Applications for which brittle materials may be used (see 3.2.3.f).

h. Sea state conditions at which the hoist is required to operate (see 3.4.3 and 4.5.5).

i. If shock test is required (see 3.4.8 and 4.5.10).

j. Rated load required (see 3.4.12 and 4.5.7).

k. If hoist characteristics are to be other than those specified in tables I, II, or III (see 3.4.12, 3.4.12.1, 3.4.12.2, 3.4.12.3, and tables I, II, and III).

l. Envelope dimensions required (see 3.4.12).

m. Type of lifting medium (link chain, roller chain, or wire rope) required (see 3.4.16, 3.4.16.1, 3.4.16.2, and 3.4.16.3).

n. If chain plating is required (see 3.4.16.1 and 3.4.27.2).

o. If chain basket, bag, or reel is required (see 3.4.16.4).

p. Type of load hook required (see 3.4.17).

q. If hook plating is required (see 3.4.17.1).

r. If pendant control is required (see 3.4.20.1).

s. Length of air valve control pendant (see 3.4.20.1).

t. If manual operation is required (see 3.4.27).

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u. Type of manual operation required (see 3.4.27 and 3.4.28.1) as follows:
   (1) Releasing of brakes.
   (2) Raising or lowering of hook.
   (3) Traversing.

v. If manual operation chain wheels are provided, they are as specified (see 3.4.27.1).

w. Drop length of hand chain required for the following:
   (1) Manual operation of hoist (see 3.4.27.2).
   (2) Operation of geared trolley (see 3.4.28.1).
   (3) Operation of trolley track clamp (see 3.4.28.6).

x. Size of I-beam support for trolley hoist, including nominal width of beam flange (see 3.4.28.1).

y. If track clamps are required (see 3.4.28.6).

z. If trolley wheels are not to be spaced to negotiate curved track (see 3.4.28.7).

aa. Color of the top coat of paint (see 3.4.29).

bb. If more than one first article sample is required (see 4.2).

cc. Frequency of conformance inspection (see 4.3).

dd. Requirements for record keeping (see 4.3.2 and 6.6).

ee. Requirements for addressing failure of the endurance test (see 4.5.6).

ff. Packaging requirements (see 5.1).

6.3 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, they must be acquired under a separate contract line item in the contract.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be preproduction sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2. The contracting officer should also include specific Technical Authority approval of first article tests and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract (see 3.1). Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Shock tests. When an identical hoist conforming to the requirements of this specification has been successfully shock tested, the test report may be submitted to the contracting activity as fulfillment of the requirements for shock testing.

6.6 Record keeping. Records of inspections, tests, and any resulting rejections should be maintained for a period of 3 years after contract completion.

6.7 Definitions.

6.7.1 Brittle material. Material showing less than 10 percent elongation in gauge length for the tensile test specimen. For the static loading case (normal ship operation), brittle material is material used below its nil ductility transition (NDT) temperature as measured by the Naval Research Laboratory (NRL) drop weight test. Charpy V-notch impact strength values may be used as criteria when these have been correlated with drop weight test results.

6.7.2 Complete cycle. A complete cycle of operation is defined as hoisting and lowering a rated load through the full hoisting range at the specified hoisting and lowering speeds (see 3.4.4, 3.7.2, 4.5.6, 4.5.7, 4.5.9.2, and 4.5.34.2).

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6.7.3 Excessive wear. Wear that is sufficient to impair safe operation of the hoist or wear that exceeds 0.031 inch.

6.7.4 Headroom. With the wire rope or chain in the fully retracted position and the load hook carrying full load, headroom is the distance between the saddle of the load hook and the following points:

a. The bottom of the lug hole on lug-suspended hoists.

b. The bottom of the beam rail on trolley-suspended hoists.

c. The saddle of the top hook on hook-suspended hoists.

6.8 Subject term (key word) listing.

Geared trolley suspension
Hand chain
Hook suspension
Load chain
Plain trolley suspension
Roller chain

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH
Air Force – 84

Preparing activity:
Navy – SH
(Project 3950-2023-002)

Review activities:
Army – GL
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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